

Homework Assignment 9
 Econ 629 - Texas A&M University
 Fall 2012

Posted: Thursday, Nov. 1st, 2012.

Due: Tuesday, Nov. 6th (Class time).

1. Problems 7, 8 in Section 6.2.1. in Lecture notes.
2. Let $Z \equiv \{a, b, c\}$ be a set of three prizes. Consider an agent who has complete and transitive preferences on the set of lotteries on Z , i.e., $\Delta(Z)$. The generic lottery is denoted by $p \equiv (p_a, p_b, p_c)$ where p_a is the probability that the agent receives a , and so on.

- (a) State the Independence axiom for a preference on the set of lotteries $\Delta(Z)$.
- (b) Assume that the agent's preferences are continuous and satisfy the independence axiom. Suppose that she selects $\{(1, 0, 0)\}$ as the set of her best choices from the set

$$B_1 \equiv \left\{ (1, 0, 0), \left(0, \frac{1}{2}, \frac{1}{2}\right) \right\}.$$

What is her best choice from the set

$$B_2 \equiv \left\{ \left(\alpha, \frac{1-\alpha}{2}, \frac{1-\alpha}{2} \right) : \alpha \in [0, 1] \right\}?$$

- (c) Now, suppose that the agent's indifference curves are as shown in Figure 1 (there is no relation, in principle, between the preferences described in part (b) and part (c); dashed lines are construction lines). Is this agent a risk averse Expected Utility maximizer?

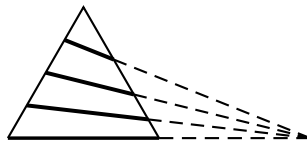


Figure 1: $\Delta(Z)$ when $Z \equiv \{a, b, c\}$.